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FACULTY OF SCIENCES
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LMD L1
2025/2026

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Algebra I

Worksheet n° : 2

Sets, Mappings & Binary relations

Exercise (01): Let the following parts of $\mathbb{R}$ be given:

$$A_1 = \{1, 2, 3\}, A_2 = [1, 3[, A_3 =]2, 4], A_4 = \{0, 1, 2, 3\}.$$

1. Calculate: $A_1 \cup A_2$, $A_2 \cup A_3$, $A_1 \cup A_4$, $A_2 \cup A_4$, $A_1 \cap A_2$, $A_2 \cap A_3$, $(A_3 \cap A_4) \cup (A_1 \cap A_4)$, $\overline{A_1}$, $\overline{A_2}$, $A_1 \times A_2$, $A_2 \times A_3$ and $A_1 \times A_4$.
2. Show that $A_1 \times A_2 \neq A_2 \times A_1$, then determine $(A_1 \times A_2) \cap (A_2 \times A_1)$.
3. Determine the elements of the following sets: $E_i = \{x \in A_i / x^2 - 7x + 10 \leq 0\}$ for $i \in \{1, 2, 3, 4\}$ and $F = \{x \in \mathbb{Z} / 2x^2 - 4 \geq 0\}$.

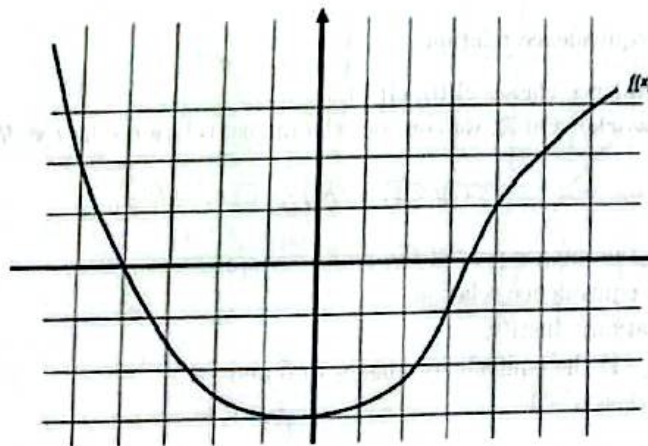
Exercise (02):

1. Let A and B be parts of a set E . Show that: $\overline{(A \cup B)} = (\bar{A} \cap \bar{B})$.
2. Give choices for the sets $A, B \subseteq \mathbb{R}$, such that:

$$\forall x \in A, \exists! y \in B, 2x - y^2 = 4.$$

3. Let $E = \mathbb{R}^2$ and $A = \{(x, y) / x^2 + y^2 = 1\}$. Prove that A cannot be written as a cartesian product of two parts of $\mathbb{R}$.

Exercise (03): Let f be a mapping from $\mathbb{R}$ to $\mathbb{R}$ whose graph is drawn below.



1. Calculate $f(\{-4, 0, 3, 4\})$, $f^{-1}(\{2\})$.
2. Calculate $f([0, 4])$, $f([-4, 4])$, $f^{-1}([1, 2])$.

3. Is the application f injective? surjective?

Exercise (04 Homework):

We consider the application $f : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto |x| - 1$ and let $A = [-1, 4]$. Determine

- The direct image of A by f .
- The inverse image of A by f .

Exercise (05): Let f be the mapping defined from $\mathbb{R}_+^*$ to $\mathbb{R}$

$$f(x) = \ln \left(\frac{x}{x+1} \right).$$

1. Determine the following sets: $f(\{\frac{1}{2}, 1, \frac{1}{e}\})$; $f([1, e])$; $f^{-1}(\{-\ln(2), 0\})$ and $f^{-1}(\mathbb{R}_+)$.
2. Show that f is injective on $\mathbb{R}_+^*$.
3. Show that f is bijective from $\mathbb{R}_+^*$ to $]-\infty, 0[$, then give its inverse f^{-1} .

Exercise (06): Are the following applications injectives? surjectives? bijectives? Determine the inverse of the application when possible.

$$1- f : \mathbb{R}^2 \rightarrow \mathbb{R}^2; (x, y) \mapsto (2x + y, 3x - y), \quad 2- g : \mathbb{R} \rightarrow \mathbb{R}_+^*; x \mapsto \frac{e^x + 2}{e^x - x}.$$

Exercise (07): On $\mathbb{R}^2$, we consider the binary relation noted $\mathcal{R}$ and defined by:

$$\forall (a, b), (c, d) \in \mathbb{R}^2 : (a, b) \mathcal{R} (c, d) \Leftrightarrow a - 5d = c - 5b.$$

1. Check that $\mathcal{R}$ is an equivalence relation.
2. Determine $Cl(0, 0)$ and $Cl(1, 1)$ the equivalence classes of the pairs $(0, 0)$ and $(1, 1)$.

Exercise (08): Show that the following binary relation on $I =]1, +\infty[$ is a total order relation:

$$\forall x, y \in I : x \mathcal{R} y \Leftrightarrow \frac{x}{1+x^2} \leq \frac{y}{1+y^2}.$$

Exercise (09): We define the following binary relation on $\mathbb{R}$

$$\forall x, y \in \mathbb{R}, \quad x \mathcal{R} y \Leftrightarrow \cos^2 x + \sin^2 y = 1.$$

1. Show that $\mathcal{R}$ is an equivalence relation.
2. Determine the equivalence classes $cl(0)$, $cl(\frac{\pi}{2})$.

Exercise (10 Homework): On $\mathbb{R}$, we consider the binary relation denoted $\mathcal{R}$ and defined by:

$$\forall x, y \in \mathbb{R} : x \mathcal{R} y \Leftrightarrow E(x) = E(y),$$

where $E(x)$ represents the integer part of the real x .

1. Check that $\mathcal{R}$ is an equivalence relation.
2. $\mathcal{R}$ is it an order relation? Justify.
3. Find $Cl(0)$ and $Cl(-1)$ the equivalence classes of 0 and -1.
4. Determine $Cl(\frac{1}{a})$, when $a > 1$.